

10. Multiple-Choice (10 Points) Medium

已知 $\Omega: x^2 + y^2 + z^2 \leq 1$, 则三重积分

$$\iiint_{\Omega} (2x + xy + z^2) dV =$$

$$\text{原式} = \iiint_{\Omega} z^2 dV$$

$$= \int_{-1}^1 dz \int_{D_z} z^2 dx dy$$

$$D_z = \{(x, y) | x^2 + y^2 \leq 1 - z^2\}$$

$$\int_{D_z} z^2 dx dy = z^2 \pi (1 - z^2)$$

$$\pi \int_{-1}^1 z^2 (1 - z^2) dz = \pi \left(\frac{z^3}{3} - \frac{z^5}{5} \right) \Big|_{-1}^1$$

$$= \pi \left(\frac{1}{3} - \frac{1}{5} \right) - \pi \left(-\frac{1}{3} + \frac{1}{5} \right) = \frac{4}{15} \pi$$

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< Close 2021-2022 学年 春夏学期 微积分甲 II 第二次小测

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若函数 $f(x, y)$ 在点 $P(0,1)$ 处连续, 且满足

$$f(x, y) = 5 + x - y + x^2 + 2x(y-1) + (y-1)^2 + o(x^2 + (y-1)^2)$$

则: 下列结论中正确的是

A $f(0,1) = 4$ ✓

B $df|_{(0,1)} = dx - dy$ ✓

C $f(x, y)$ 在点 $(0,1)$ 处有极值 ✗

D $f(x, y)$ 在点 $(0,1)$ 处的切平面为 $z = x - y + 5$ ✗

$$f(0,1) = 5 - 1 + 0 + 0 = 4$$

$$df = dx - dy + 2x dx + 2x dy + 2(y-1) dx + 2(y-1) dy$$

$$= (2x + 2y - 1) dx + (2x + 2y - 3) dy$$

$$\frac{\partial f}{\partial x} = 0 = \frac{\partial f}{\partial y}$$

$$x + y = \frac{1}{2}$$

$$x + y = \frac{3}{2}$$

$$[(y-1) + x]^2 + x - (y-1) + 4$$

$$(a+b)^2 + a - b + 4 \text{ 极值}$$

$$\begin{cases} \frac{\partial z}{\partial x} = 1 \\ \frac{\partial z}{\partial y} = -1 \\ \frac{\partial z}{\partial z} = 1 \end{cases} \quad \begin{cases} 1(x-0) - 1(y-1) + 1(z-4) = 0 \\ x - y + 1 + z - 4 = 0 \\ z = 3 \end{cases}$$

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7. Multiple-Choice (10 Points) Medium

在空间曲线 $r(t) = \{x(t), y(t), z(t)\} = \{t, t^2, t^3\}$ 的所有切线中, 与平面 $x + 2y + z = 4$ 平行的切线

A 只有两条

B 只有一条 ✗

C 不存在 ✗

D 至少有三条

$$\text{切线方向 } \vec{n} = \{1, 2t, 3t^2\}$$

$$\vec{n}_2 = \{1, 2, 1\}$$

$$A \quad \begin{cases} 1 - 4t + 3t^2 = 0 \\ (3t-1)(t-1) = 0 \end{cases}$$

3. 单选题 (10 分) 中

设 $f(x, y)$ 在 $(0, 0)$ 处连续, 且

$\lim_{(x,y) \rightarrow (0,0)} \frac{f(x,y) - xy}{(x^2 + y^2)^2} = 1$, 则 $(0, 0)$ 是 $f(x, y)$ 的
() .

A.

A | 不是极值点

B | 极小值点

C | 极大值点

D | 不能判断是否极值点.

$$\lim_{(x,y) \rightarrow (0,0)} f(x,y) = (x^2 + y^2)^2 + xy$$

$$\frac{\partial f}{\partial x} = 2(x^2 + y^2) \cdot 2x + y$$

$$\frac{\partial^2 f}{\partial x^2} = 12x^2 + 4y^2$$

$$\frac{\partial^2 f}{\partial y^2} = 0$$

$$\frac{\partial f}{\partial x \partial y} = 1 + 8xy = 1$$

$AC - B^2 < 0$. 不是.

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8. Multiple-Choice (10 Points) Medium

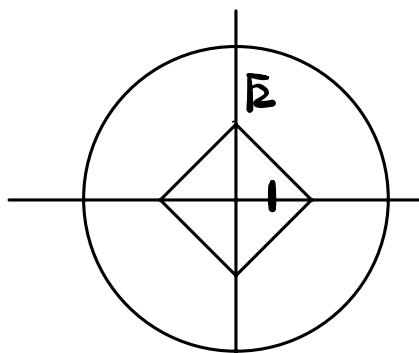
已知 $D: \{(x, y) | x^2 + y^2 \leq 2, |x| + |y| \geq 1\}$,

则: $\iint_D (x+y)^2 dx dy =$

A |

$$2\pi - \frac{2}{3}$$

▲ ▼ next



$$y = -x + 1$$

$$r \sin \theta = -r \cos \theta$$

$$r =$$

$$\begin{aligned} & \iint_D (x^2 + y^2) dx dy \\ &= \int_0^{2\pi} d\theta \int_{\frac{1}{\sin\theta + \cos\theta}}^{\sqrt{2}} r^2 \cdot r dr \\ &= \int_0^{2\pi} \left(1 - \frac{1}{4(\sin\theta + \cos\theta)^4} \right) d\theta \\ &= 2\pi - \frac{1}{4} \int_0^{2\pi} \frac{1}{(\sin\theta + \cos\theta)^4} d\theta \end{aligned}$$

9.多选题(10分) 中

设函数 $z = f(x, y)$ 在 (x_0, y_0) 处可微, 则有().

A $f(x, y)$ 在 (x_0, y_0) 处一阶偏导数存在. ✓

B $f(x, y)$ 在 (x_0, y_0) 处连续. ✓

C $f(x, y)$ 在 (x_0, y_0) 处沿任意方向的方向导数存在. ✓

D 若 $\frac{\partial z}{\partial x}(x, y)$ 在 (x_0, y_0) 的某个邻域内有定义, 则 $\frac{\partial z}{\partial x}(x, y)$ 在 (x_0, y_0) 处连续. ✗

极值的必要条件.

(x_0, y_0) 可微且取极值

$$\Rightarrow f'_x(x_0, y_0) = f'_y(x_0, y_0) = 0$$

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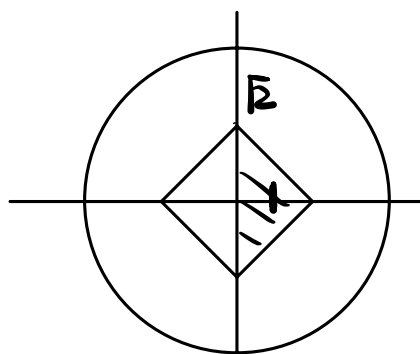
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8. Multiple-Choice (10 Points) Medium

已知 $D: \{(x, y) | x^2 + y^2 \leq 2, |x| + |y| \geq 1\}$,

则: $\iint_D (x+y)^2 dx dy =$

A $2\pi - \frac{2}{3}$



增补法

$$\iint_{D_1} (x^2 + y^2) dx dy$$

$$= \int_0^{2\pi} d\theta \int_0^{\sqrt{2}} r^2 \cdot r dr = \int_0^{2\pi} 1 d\theta = 2\pi$$

$$\iint_{D_2} (x^2 + y^2) dx dy = 2 \iint_{D_2} x^2 dx dy$$

$$= 4 \int_0^1 \int_0^{1-x} x^2 dx dy = 4 \int_0^1 dx \int_{x-1}^{1-x} x^2 dy$$

$$= 4 \int_0^1 x^2 (2 - 2x) dx = \frac{2}{3}$$

$$\therefore 2\pi - \frac{2}{3}$$

10. 已知 $D = \{(x, y) | 0 \leq x \leq 1, 0 \leq y \leq 1\}$,

则: $\iint_D e^{\max\{x, y\}} dx dy =$

单选题 (10 分) (难度: 中)

- ☐ A. e
- ☐ B. 2
- ☐ C. $2e-1$
- ☐ D. $e+1$

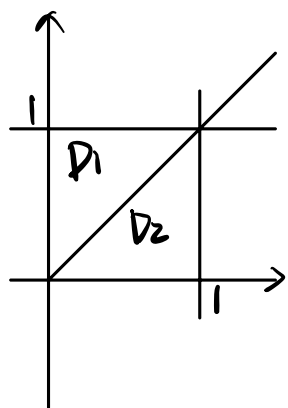
答题数据分析

答对: 150

答错: 70

未答: 14

正确率: 68.18%



$$\begin{aligned} & \iint_{D_1} e^y dx dy + \iint_{D_2} e^x dx dy \\ &= \int_0^1 dy \int_0^y e^y dx + \int_0^1 dx \int_0^x e^x dy \\ &= \int_0^1 y e^y dy + \int_0^1 x e^x dx \\ &= 2 \int_0^1 x e^x dx = 2 \int_0^1 x d e^x \\ &= 2 x e^x \Big|_0^1 - 2 \int_0^1 e^x dx \\ &= 2 x e^x \Big|_0^1 - 2 e^x \Big|_0^1 \\ &= 2e - (2e - 2) = 2 \end{aligned}$$

23春夏第二次随堂小测

2/10

23春夏第二次随堂小测

5/10

2.单选题(10分)

设 $f(x, y)$ 在 $\mathbb{R}^2$ 上具有连续偏导数, 且 $f(1,1)=1$
 $f'_x(1,1)=1, f'_y(1,1)=2$. 记 $\varphi(x)=f(x, f(x, x^2))$.
 则: $\varphi'(1)=$

A 5;

B 11. ✓

C 7;

D 9;

5.单选题(10分)

设 $\Omega: x^2+y^2+z^2 \leq z$. 则: 三重积分 $\iiint_{\Omega} (x+y+z) dV =$

A $\frac{\pi}{6}$.

B $\frac{\pi}{24}$.

C $\frac{\pi}{4}$.

D $\frac{\pi}{12}$. ✓

23春夏第二次随堂小测

6/10

6.单选题(10分)

设 $f(x, y) = \begin{cases} \frac{x^2 y}{x^2 + y^2} & (x^2 + y^2 \neq 0) \\ 0 & (x^2 + y^2 = 0) \end{cases}$. 给定方向 $l = (-1, 1)$.
 则: 函数 $f(x, y)$ 在 $(0,0)$ 处沿 l 方向的方向导数 $\frac{\partial f}{\partial l} \Big|_{(0,0)} =$

A -1;

B -1/2;

C $\frac{\sqrt{2}}{4}$. ✓

D 0;

23春夏第二次随堂小测

10/10

10.多选题(10分)

$2x^2 + 2y^2 + z^2 + 8xz - z + 8 = 0$
 方程所确定的隐函数 $z = f(x, y)$

A 不存在极大值

B 存在极小值 ✓

C 存在极大值 ✓

D 不存在极小值

23春夏第二次随堂小测

3/10

3.单选题(10分)

在空间曲线 $r(t) = \{x(t), y(t), z(t)\} = \{e-t^2, t^3\}$ 的所有切线中, 与平面 $12x + 3y - 2z = 4$ 平行的切线

A 不存在

B 只有一条

C 只有两条 ✓

D 至少有三条

定义: $\rho \vec{p} = \rho \frac{\vec{l}}{|\vec{l}|} = \left\{ \frac{\rho}{\sqrt{2}}, \frac{\rho}{\sqrt{2}} \right\}$

$\rho \left(\frac{\rho}{\sqrt{2}}, \frac{\rho}{\sqrt{2}} \right)$

$$\frac{\partial f}{\partial l} \Big|_{(0,0)} = \lim_{\rho \rightarrow 0} \frac{f(\rho \vec{p}) - f(0)}{\rho} = \frac{\left(\frac{\rho}{\sqrt{2}} \right)^2 \frac{\rho}{\sqrt{2}}}{\left(\frac{\rho}{\sqrt{2}} \right)^2 + \left(\frac{\rho}{\sqrt{2}} \right)^2} \times \frac{1}{\rho}$$

$$= \frac{\frac{1}{2\sqrt{2}} \cdot \rho^3}{\rho^2} = \frac{\sqrt{2}}{4}$$

$$\begin{cases} \psi x + 2z \frac{\partial z}{\partial x} + 8x \frac{\partial z}{\partial x} + 12z - \frac{\partial z}{\partial x} = 0 \\ \psi y + 2z \frac{\partial z}{\partial y} + 8x \frac{\partial z}{\partial y} - \frac{\partial z}{\partial y} = 0 \end{cases} \Rightarrow \begin{cases} x + 2z = 0 \\ y = 0 \end{cases} \Rightarrow (2, 0) \left(\frac{1}{\sqrt{2}}, 0 \right)$$

$$\begin{cases} \psi + 2z \frac{\partial^2 z}{\partial x^2} + 2 \left(\frac{\partial z}{\partial x} \right)^2 + 8x \frac{\partial^2 z}{\partial x^2} + 12 \left(\frac{\partial z}{\partial x} \right)^2 + 8 \frac{\partial^2 z}{\partial x^2} - \frac{\partial^2 z}{\partial x^2} = 0 \\ 2z \frac{\partial^2 z}{\partial x \partial y} + 8x \frac{\partial^2 z}{\partial x \partial y} + 8 \frac{\partial^2 z}{\partial y^2} - \frac{\partial^2 z}{\partial x \partial y} = 0 \end{cases} \Rightarrow \begin{cases} A = \frac{\psi}{12x-2z} \\ B = 0 \\ C = \frac{\psi}{12x-2z} \end{cases}$$

$$\psi + 2z \frac{\partial^2 z}{\partial y^2} + 8x \frac{\partial^2 z}{\partial y^2} - \frac{\partial^2 z}{\partial y^2} = 0$$

$B^2 - AC < 0$ 有极值.

$A = \frac{\psi}{12x-2z} = \begin{cases} + \\ - \end{cases}$

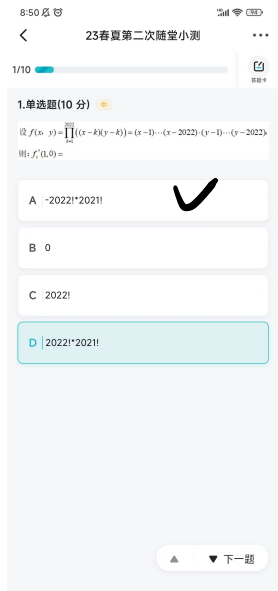
切向量 $\vec{l} = \{1, -2t, 3t^2\}$

$\vec{n} = \{12, 3, -2\}$

$12 - 6t - 6t^2 = 0$

$t^2 + t - 2 = 0$

$(t+2)(t-1) = 0$

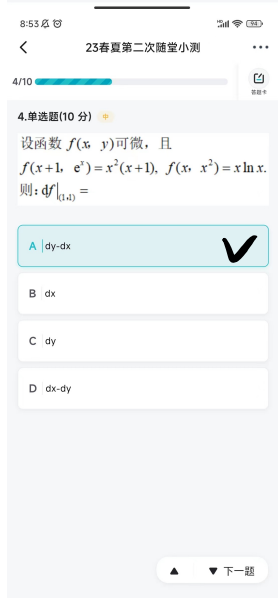


$$H) x(2) x \cdot (-2022) = 2022!$$

$$\lim_{x \rightarrow 1} \frac{f(x,0) - f(1,0)}{x-1} = \lim_{\Delta x \rightarrow 0} \frac{f(1+\Delta x, 0) - f(1,0)}{\Delta x} = 2022! \lim_{\Delta x \rightarrow 0} \frac{\Delta x (\Delta x - 1) \cdots (\Delta x - 2022)}{\Delta x}$$

$$= 2022! (-1)(-2) \cdots (-2021)$$

$$= -2022! \cdot 2021!$$



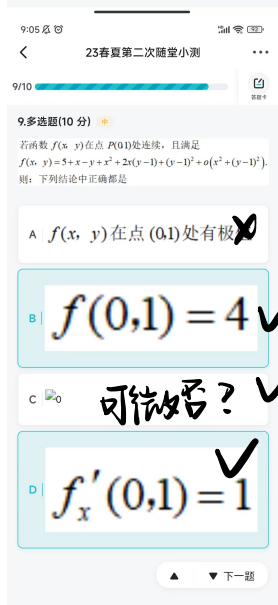
$$f'_x + f'_y \cdot e^x = 3x^2 + 2x \Rightarrow f'_x + f'_y = 0$$

$$f'_x + f'_y \cdot 2x = 1 + \ln x \Rightarrow f'_x + 2f'_y = 1$$

$$\Rightarrow f'_y = 1, f'_x = -1$$

$$\begin{cases} f'_x(x+1, e^x) + e^x f'_y(x+1, e^x) = 3x^2 + 2x \\ f'_x(x, x^2) + 2x f'_y(x, x^2) = \ln x + 1 \end{cases}$$

$$\Rightarrow \begin{cases} f'_x + f'_y = 0 \\ f'_x + 2f'_y = 1 \end{cases} \Rightarrow \begin{cases} f'_x = -1 \\ f'_y = 1 \end{cases}$$



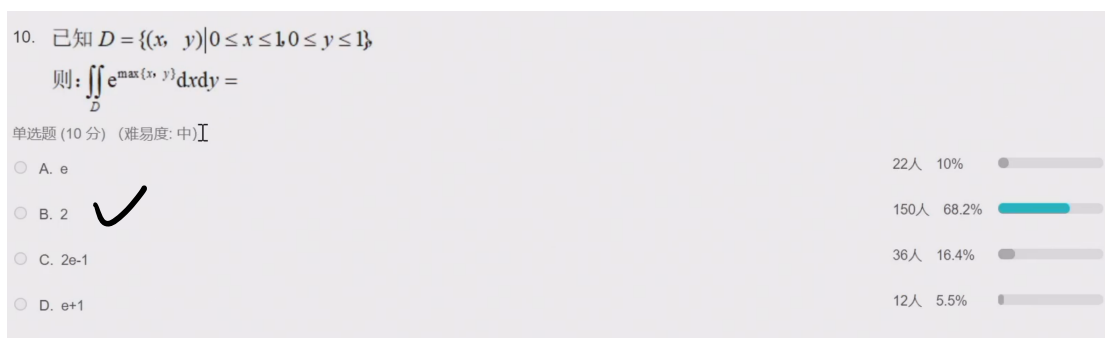
$$f(0,1) = 5 - 1 = 4$$

$$\frac{\partial f}{\partial x} = 1 + 2x + 2y - 2 = 2x + 2y - 1$$

$$\frac{\partial f}{\partial y} = -1 + 2x + 2(y-1) = 2x + 2y - 3$$

有偏导连续

可微否? ✓✓



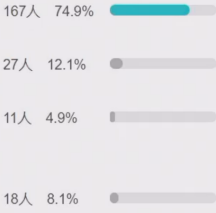
9. 曲面 $e^{2z} - xyz = e^2$ 在点 $M(0, 2, 1)$

处的切平面方程

单选题 (10 分) (难度: 中)

- ☐ A. $e^{-2}x - z + 1 = 0$
- ☐ B. $e^2x - z + 1 = 0$
- ☐ C. $\frac{2}{2e^2 + 1}x - z + 1 = 0$
- ☐ D. $\frac{2}{2e^2 - 1}x - z + 1 = 0$

$$F(x, y, z) = e^{2z} - xyz - e^2 = 0$$
$$F_x' = -yz = -2$$
$$F_y' = -xz = 0$$
$$F_z' = 2e^{2z} - xy = 2e^2$$
$$-2(x-0) + 2e^2(y-2) + 2e^2(z-1) = 0$$
$$x - 2e^2(z-1) = 0$$



答题数据分析 答对: 167 答错: 56 未答: 11 正确率: 74.89%

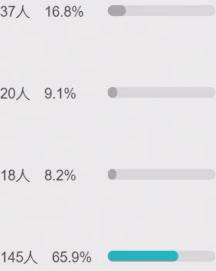
8. 已知 $\Omega: x^2 + y^2 + z^2 \leq 1$, 则三重积分

$$\iiint_{\Omega} (2x + xy + z^2) dV =$$

单选题 (10 分) (难度: 中)

- ☐ A. $\frac{19}{15}\pi$
- ☐ B. $\frac{8}{5}\pi$
- ☐ C. $\frac{23}{30}\pi$
- ☒ D. $\frac{4}{15}\pi$

$$\iiint_{\Omega} z^2 dV$$
$$= \int_{-1}^1 dz \int_0^{2\pi} d\theta \int_0^1 r^2 dr$$
$$= \int_{-1}^1 z^2 \pi r^2 dz$$
$$= \pi \left[\frac{z^3}{3} - \frac{z^5}{5} \right]_{-1}^1 = \frac{4}{15}\pi$$



答题数据分析 答对: 145 答错: 75 未答: 14 正确率: 65.91%

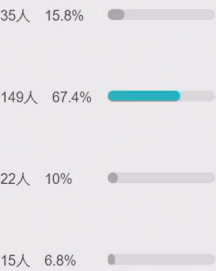
7. 已知 $D: \{(x, y) | x^2 + y^2 \leq 2, |x| + |y| \geq 1\}$,

则: $\iint_D (x + y)^2 dx dy =$

单选题 (10 分) (难度: 中)

- ☐ A. $2\pi - \frac{3}{4}$
- ☒ B. $2\pi - \frac{2}{3}$
- ☐ C. $\pi - \frac{2}{3}$
- ☐ D. $\pi + \frac{4}{3}$

$$f(0, 1) = 5 - 1$$
$$f_x' = 1 + 2x + 2y - 2 = 2x + 2y - 1$$
$$f_y' = 1 + 2x + 2y - 1 = 2x + 2y - 3$$
$$f(0, 1) = 1$$
$$f(x, y) - z = F(x, y, z) = 0$$
$$F_x' = f_x' = 1, F_y' = f_y' = -1, F_z' = -1$$



答题数据分析 答对: 149 答错: 72 未答: 13 正确率: 67.42%

6. 若函数 $f(x, y)$ 在点 $P(0, 1)$ 处连续, 且满足

$$f(x, y) = 5 + x - y + x^2 + 2x(y - 1) + (y - 1)^2 + o(x^2 + (y - 1)^2).$$

则: 下列结论中正确的是

多选题 (10 分) (难度: 中)

- ☒ A. $f(0, 1) = 4$
- ☐ B. $f_x'(0, 1) = 1$
- ☒ C. $df|_{(0, 1)} = dx - dy$
- ☐ D. $f(x, y)$ 在点 $(0, 1)$ 处有极值
- ☐ E. $f(x, y)$ 在点 $(0, 1)$ 处的切平面为 $z = x - y + 5$

$$f(0, 1) = 5 - 1$$
$$f_x' = 1 + 2x + 2y - 2 = 2x + 2y - 1$$
$$f_y' = 1 + 2x + 2y - 1 = 2x + 2y - 3$$
$$f(0, 1) = 1$$
$$f(x, y) - z = F(x, y, z) = 0$$
$$F_x' = f_x' = 1, F_y' = f_y' = -1, F_z' = -1$$



答题数据分析 答对: 60 答错: 152 未答: 22 正确率: 24.39%

$$1 \times (x-0) - (y-1) - (z-4) = 0$$

$$x - y - z + 5 = 0 \Rightarrow z = x - y + 5$$

1. 在空间曲线 $r(t) = \{x(t), y(t), z(t)\} = \{t-t^2, t^3\}$ 的所有切线中, 与平面 $x+2y+z=4$ 平行的切线

单选题 (10 分) (难度: 中)

- ☒ A. 只有一条
- ☐ B. 只有两条
- ☐ C. 至少有二条
- ☐ D. 不存在

$$\text{切向量 } \vec{r}' = \{1, -2t, 3t^2\}$$

$$\vec{n} = \{1, 2, 1\}$$

$$\vec{r}' \cdot \vec{n} = 1 - 2t + 3t^2 = 0$$

$$(3t-1)(t+1) = 0$$

17人 7.6%

168人 75.3%

4人 1.8%

34人 15.2%

答题数据分析 答对: 168 答错: 55 未答: 11 正确率: 75.34%

2. 设函数 $f(x, y)$ 可微, 且

$$f(x+1, e^x) = x^2(x+1), f(x, x^2) = x \ln x.$$

$$\text{则: } df|_{(1,1)} =$$

单选题 (10 分) (难度: 中)

- ☐ A. dx
- ☐ B. dy
- ☐ C. dx-dy
- ☒ D. dy-dx

28人 12.7%

52人 23.5%

47人 21.3%

94人 42.5%

答题数据分析 答对: 94 答错: 127 未答: 13 正确率: 42.53%

3. 已知 $\Omega: x^2 + y^2 + (z-1)^2 \leq 1$, 则: $\iiint_{\Omega} z dV =$

单选题 (10 分) (难度: 中)

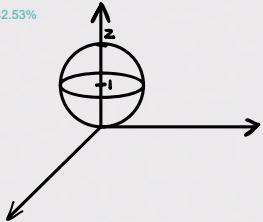
- ☐ A. $\frac{2}{3}\pi$
- ☐ B. π
- ☐ C. 2π
- ☒ D. $\frac{4}{3}\pi$

$$\int_0^1 dz \iint_{D_z} z dx dy$$

$$D_z: \{x^2 + y^2 \leq 2z - z^2\}$$

$$= \int_0^1 z \pi (2z - z^2) dz$$

$$= \pi \int_0^1 (2z^2 - z^3) dz = \pi \left[\frac{2z^3}{3} - \frac{z^4}{4} \right]_0^1 = \frac{4}{3}\pi$$



16人 7.2%

9人 4.1%

11人 5%

186人 83.8%

答题数据分析 答对: 186 答错: 36 未答: 12 正确率: 83.78%

4. 函数 $u = \sqrt{x^2 + 2y^2 + 3z^2}$ 在点 $M(2, -1, 1)$ 处的梯度假为

单选题 (10 分) (难度: 中)

- ☐ A. $\frac{\sqrt{17}}{3}$
- ☐ B. $\sqrt{17}$
- ☐ C. $\frac{1}{3}(2i - 2j + 3k)$
- ☐ D. $2i - 2j + 3k$

$$\frac{\partial u}{\partial x} = \frac{1}{2} (x^2 + 2y^2 + 3z^2)^{-\frac{1}{2}} \cdot 2x = \frac{x}{u}$$

$$\frac{\partial u}{\partial y} = \frac{1}{2} (x^2 + 2y^2 + 3z^2)^{-\frac{1}{2}} \cdot 4y = \frac{2y}{u}$$

4人 1.8%

178人 79.5%

13人 5.8%

答题数据分析 答对: 178 答错: 46 未答: 10 正确率: 79.46%

5. 设二元函数 $z = f(x, y)$ 在 $P_0(x_0, y_0)$

的某邻域内有定义, 有下面 4 个命题

$P_1: z = f(x, y)$ 在 $P_0(x_0, y_0)$ 处连续;

$P_2: z = f(x, y)$ 的两个偏导数

$f'_x(x, y), f'_y(x, y)$ 在 P_0 处连续;

$P_3: z = f(x, y)$ 在 P_0 处可微;

$P_4: z = f(x, y)$ 在 P_0 处的两个偏导数

$f'_x(x, y), f'_y(x, y)$ 存在.

则: 下列陈述正确的是

多选题 (10 分) (难度: 中)

- ☐ A. $P_2 \Rightarrow P_3 \Rightarrow P_1$
- ☐ B. $P_3 \Rightarrow P_2 \Rightarrow P_1$
- ☐ C. $P_2 \Rightarrow P_3 \Rightarrow P_4$
- ☐ D. $P_3 \Rightarrow P_1 \Rightarrow P_2$

174人 80.9%

49人 22.8%

156人 72.6%

45人 20.9%

$$P_2 \rightarrow P_3 \rightarrow P_1$$

$$\searrow P_4$$